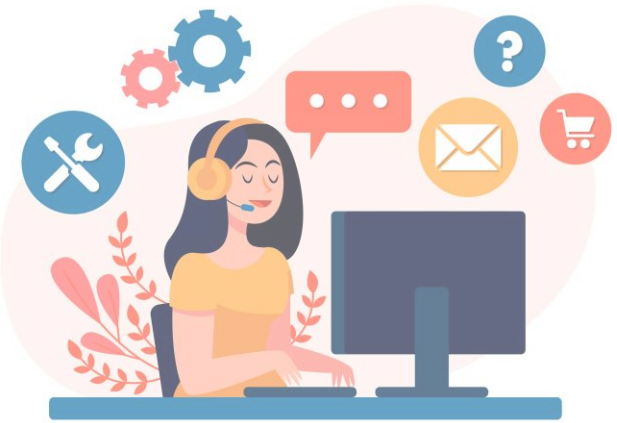
next leap



Data Analyst Fellowship Project

Flipkart

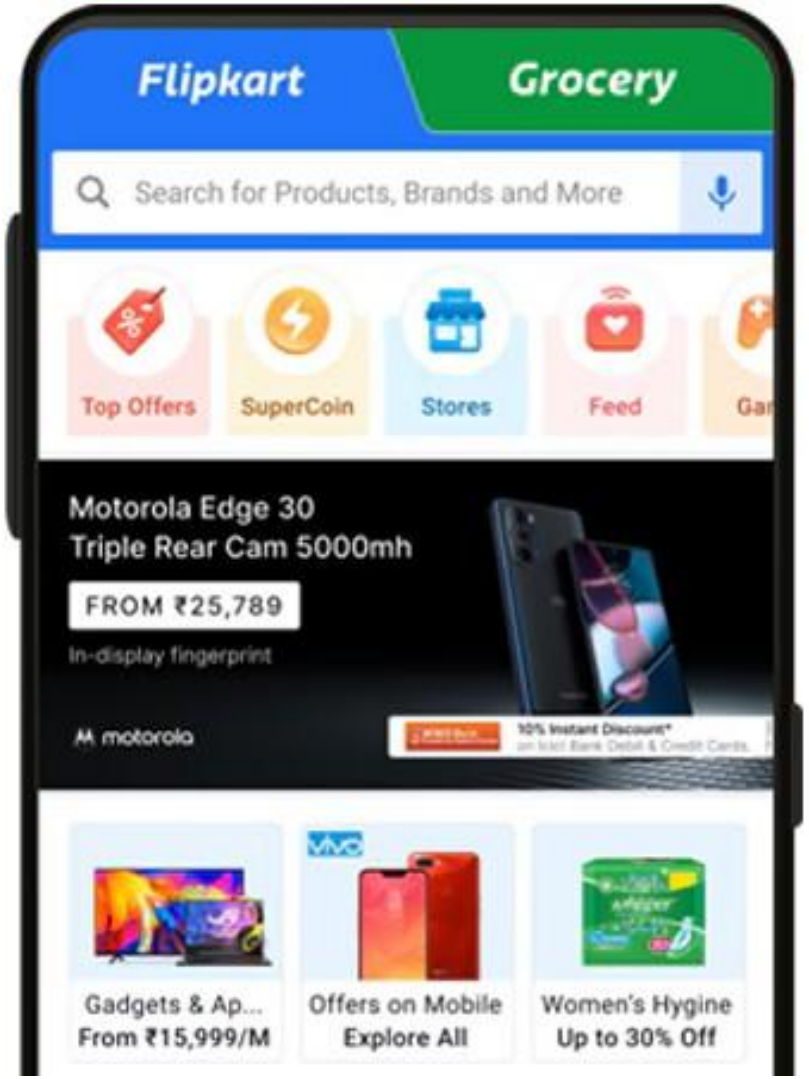
Shashwat Singh

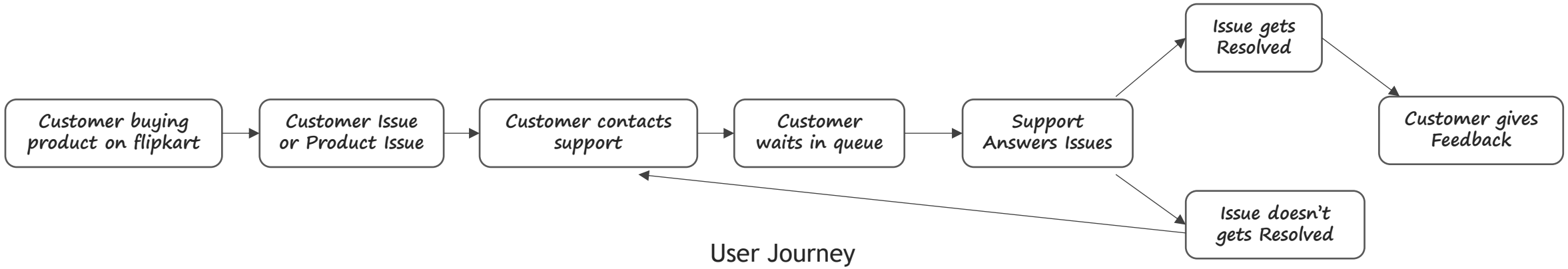
Objective

To analyse customer call data and assess various aspects of customer service performance to determine if and how it impacts customer retention.

Goal

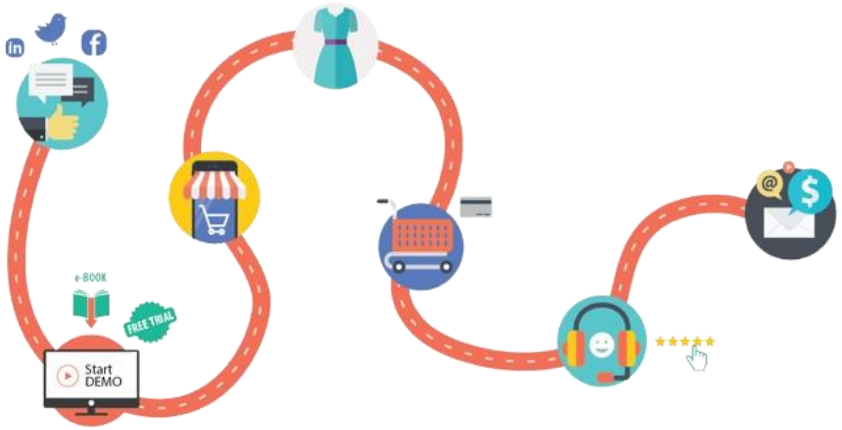
To identify specific issues within customer service operations that may be affecting customer satisfaction and retention rates.

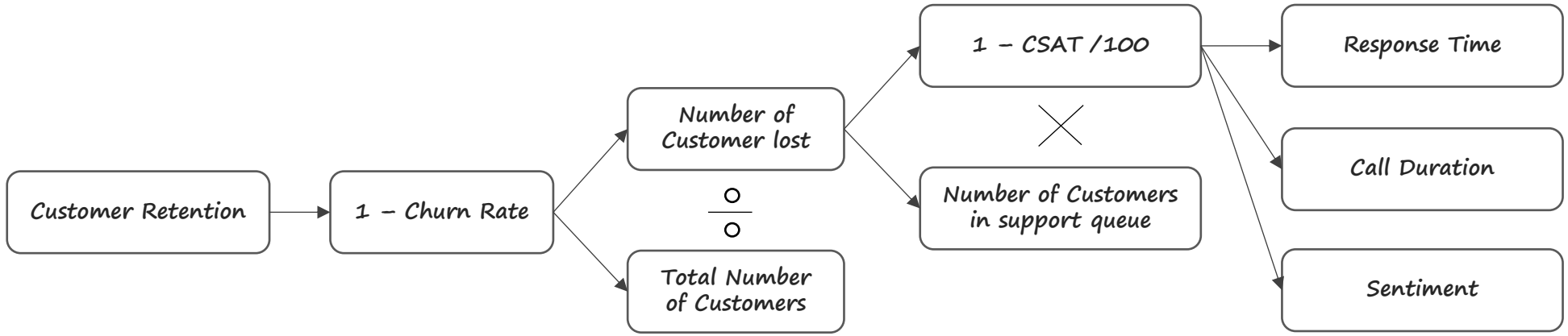




User Journey

- The customer goes on Flipkart to buy a product.
- The customer faces an issue with few things.
- The customer contacts Flipkart's support team via available channels.
- Customer service works on helping the customer by focusing on:
  - Reducing Response Time
  - First Contact Resolution
  - Improving Sentiment score and CSAT score





Metric Tree

Key Metrics

- CSAT Score: Measures how satisfied customers are with the support they receive.
- Sentiment Score: Measures the feedback rating on provided service.
- Percentage of responses within SLA: Shows the percentage of customer support responses that meet the Service Level Agreement (SLA) time limits.
- Duration to resolve issues: How quickly customer issues are resolved.
- Percentage of issues resolved in one contact: Indicates the percentage of customer issues that are resolved during the first contact with support.

Sentiment and CSAT

- There is a significant positive relationship between customer sentiment and CSAT scores.

Response Time and Call Duration Impact

- Cases with response time above SLA have lower CSAT scores.
- Cases with higher call duration have lower CSAT scores.

Support Channel Effectiveness

- Certain support channels have lower CSAT scores compared to others.

Call Centre Performance

- Certain call centers have lower CSAT scores compared to others.

- Certain types of issues are associated with lower CSAT scores.
- Within each issue type, Certain support channels have lower CSAT scores compared to others.
- Within each issue type, Certain call centers have lower CSAT scores compared to others.

Issue Type and CSAT

Regional Performance

- Customer satisfaction (CSAT) varies significantly across top states.

- CSAT scores decreased significantly throughout the month
- Certain days of the week have consistently lower CSAT scores.

CSAT trend and Day Factors

EDA Steps

- Data Import: Import the customer support data into Excel.
- Data Cleaning and Preparation: Address missing values, remove
- duplicates, and standardize data formats.
- Data Processing: Create new columns if necessary.

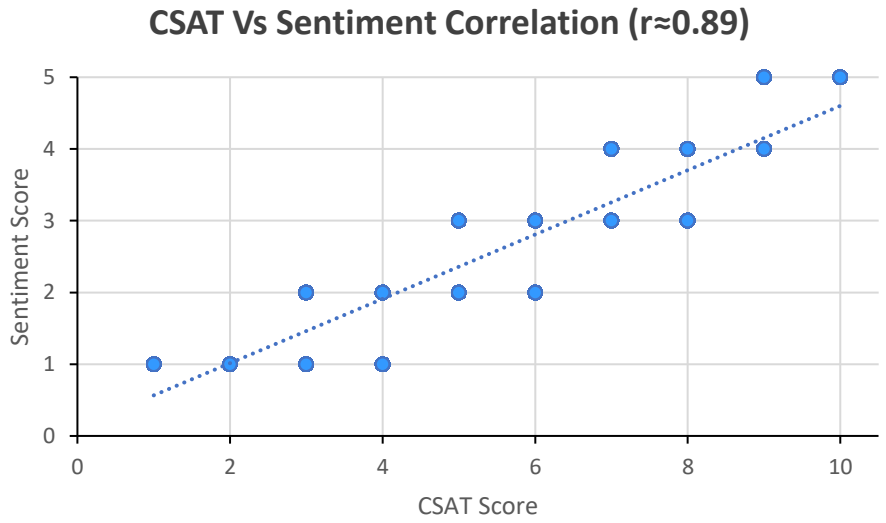
- Standardised data formats and call_timestamp column has been made uniform.
- Very few missing values are there which is not affective customer support analysis aspect.
- No duplicate values were found.
- Two new columns have been formed: 'sentiment_score', 'call_day'.

Analysis

Hypothesis:
There is a significant positive relationship between customer sentiment and CSAT scores.

Methodology:
- Conducted correlation analysis between customer sentiment and CSAT scores
- Used a scatter plot to visualize the relationship
- Added a regression trend line to identify the overall pattern
- Calculated the correlation coefficient (r) to measure strength and direction

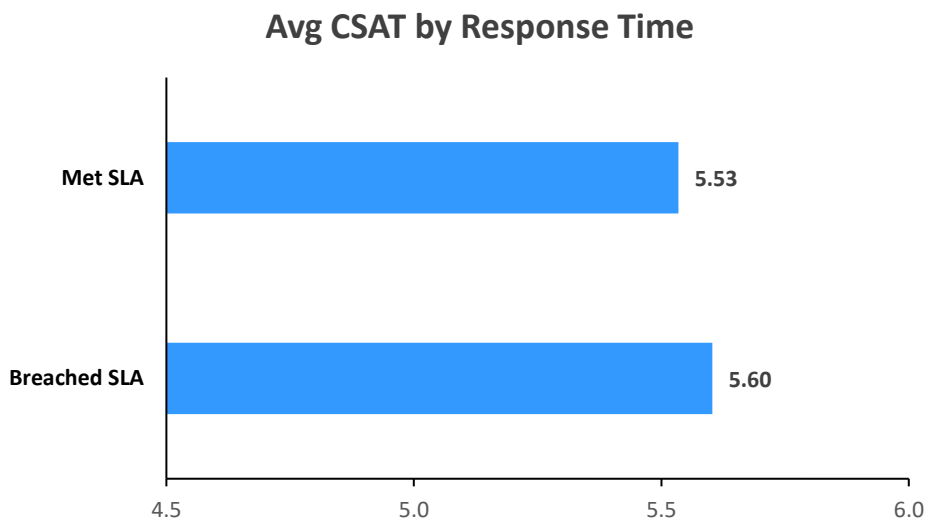
Insight:
A strong positive relationship exists between customer sentiment and CSAT scores (r = 0.89), with higher sentiment consistently associated with higher customer satisfaction



Hypothesis:
Cases with response time above SLA are expected to have lower CSAT scores.

Methodology:
- Categorized cases into Met SLA and Breached SLA
- Calculated average CSAT score for each category
- Compared CSAT performance between the two groups
- Visualized the comparison using a bar chart

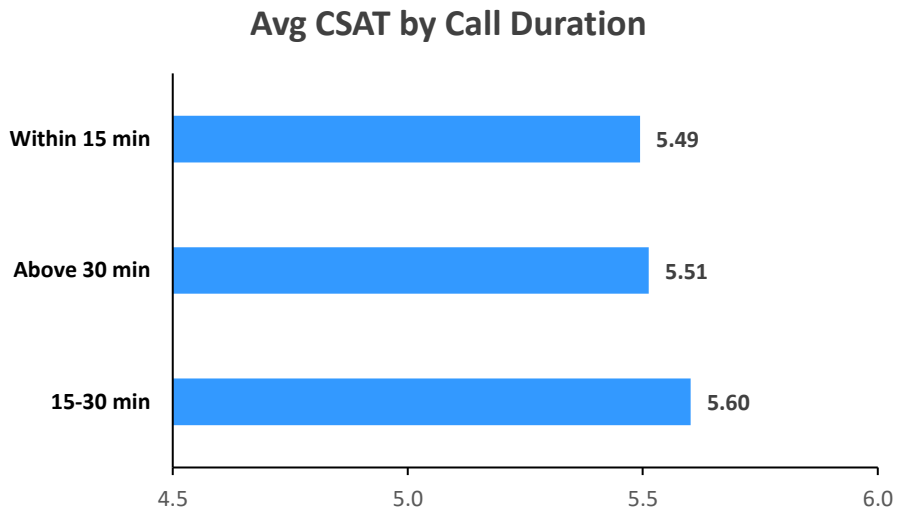
Insight:
There is no significant difference in CSAT scores between cases that met SLA (5.53) and those that breached SLA (5.60), indicating that response time does not meaningfully impact customer satisfaction



Hypothesis:
Cases with higher call duration have lower CSAT scores.

Methodology:
- Segmented cases based on call duration (Within 15 min, 15–30 min, Above 30 min)
- Calculated average CSAT score for each duration category
- Compared CSAT performance across different duration groups
- Visualized the comparison using a bar chart

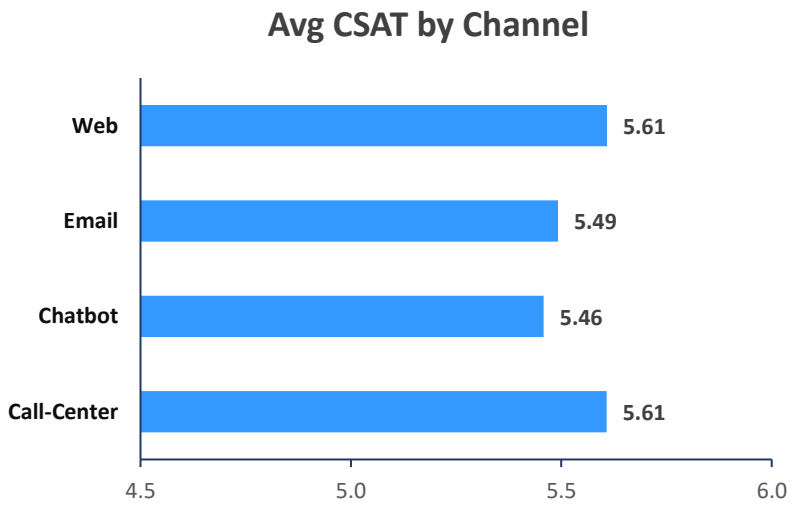
Insight:
CSAT scores show minimal variation across call duration categories (5.49–5.60), indicating no meaningful relationship between call duration and customer satisfaction



Hypothesis:
Certain support channels are expected to have lower CSAT scores compared to others.

Methodology:
- Segmented cases by support channel (Web, Email, Chatbot, Call Center)
- Calculated average CSAT score for each channel
- Compared CSAT performance across channels
- Visualized the comparison using a bar chart

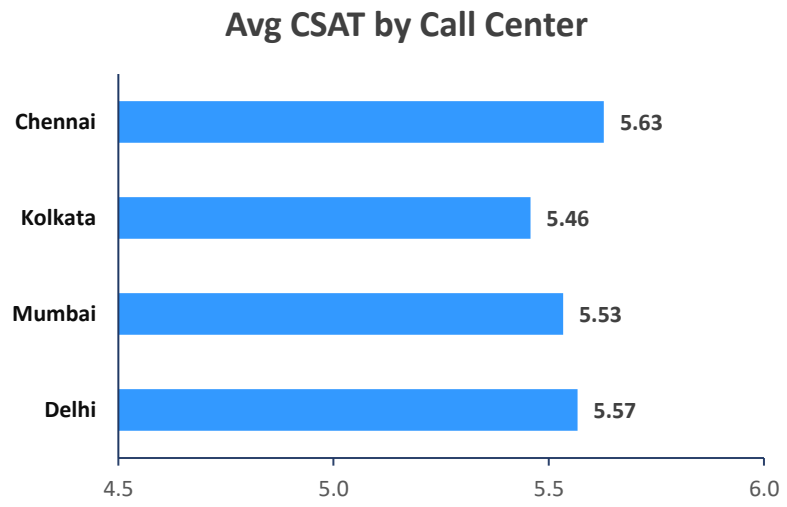
Insight:
CSAT scores show minimal variation across support channels (5.46–5.61), indicating no meaningful difference in customer satisfaction between channels



Hypothesis:
Certain call centers are expected to have lower CSAT scores compared to others.

Methodology:
- Segmented cases by call center location (Chennai, Kolkata, Mumbai, Delhi)
- Calculated average CSAT score for each call center
- Compared CSAT performance across locations
- Visualized the comparison using a bar chart

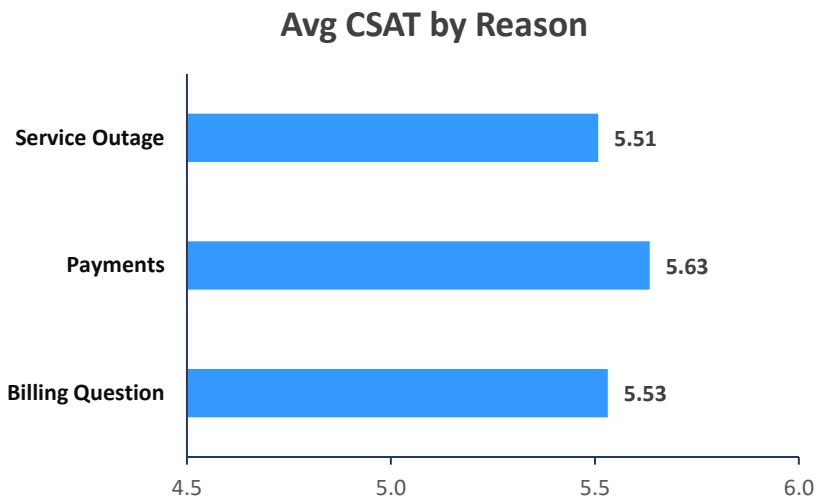
Insight:
CSAT scores show minimal variation across call centers (5.46–5.63), indicating no meaningful difference in customer satisfaction



Hypothesis:
Certain types of issues are associated with lower CSAT scores.

Methodology:
- Segmented cases by issue type (Service Outage, Payments, Billing Question)
- Calculated average CSAT score for each issue type
- Compared CSAT performance across issue categories
- Visualized the comparison using a bar chart

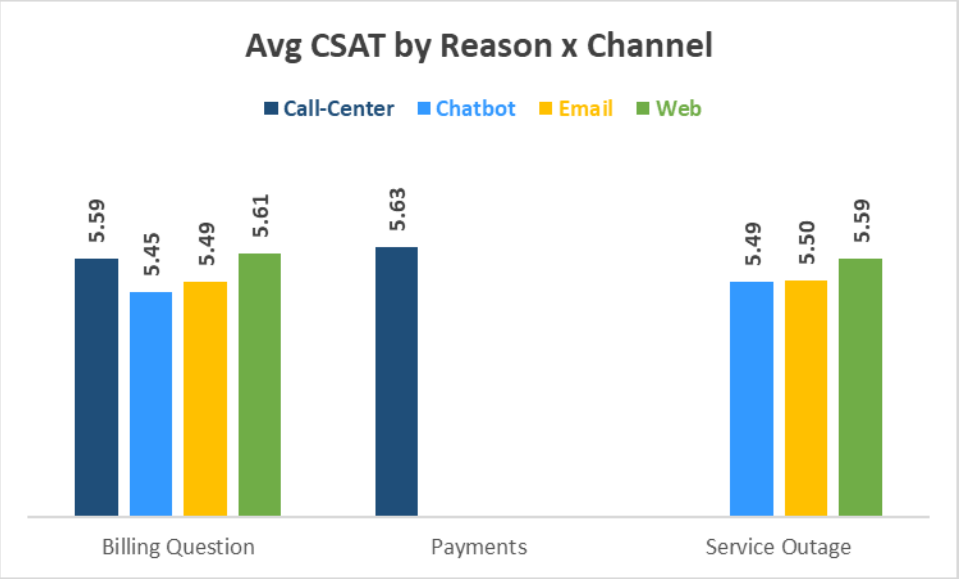
Insight:
CSAT scores show minimal variation across issue types (5.51–5.63), indicating no meaningful difference in customer satisfaction



Hypothesis:
Within each issue type, Certain support channels have lower CSAT scores compared to others.

Methodology:
- Segmented data by issue type and support channel
- Calculated average CSAT score for each combination
- Compared CSAT performance within each issue type across channels
- Visualized the comparison using a clustered bar chart

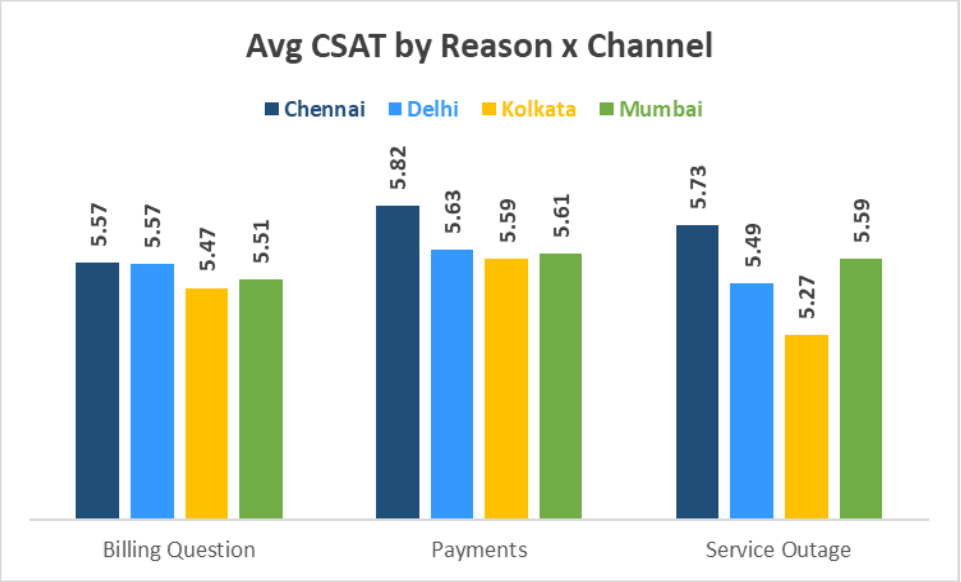
Insight:
Within each issue type, CSAT varies minimally across channels with no consistent underperformance; some channel–issue combinations have no interactions and were excluded from analysis due to lack of data.



Hypothesis:
Within each issue type, Certain call centers have lower CSAT scores compared to others.

Methodology:
•Segmented data by issue type and call center (Chennai, Delhi, Kolkata, Mumbai)
•Calculated average CSAT score for each combination
•Compared CSAT performance within each issue type across call centers
•Visualized the comparison using a clustered bar chart

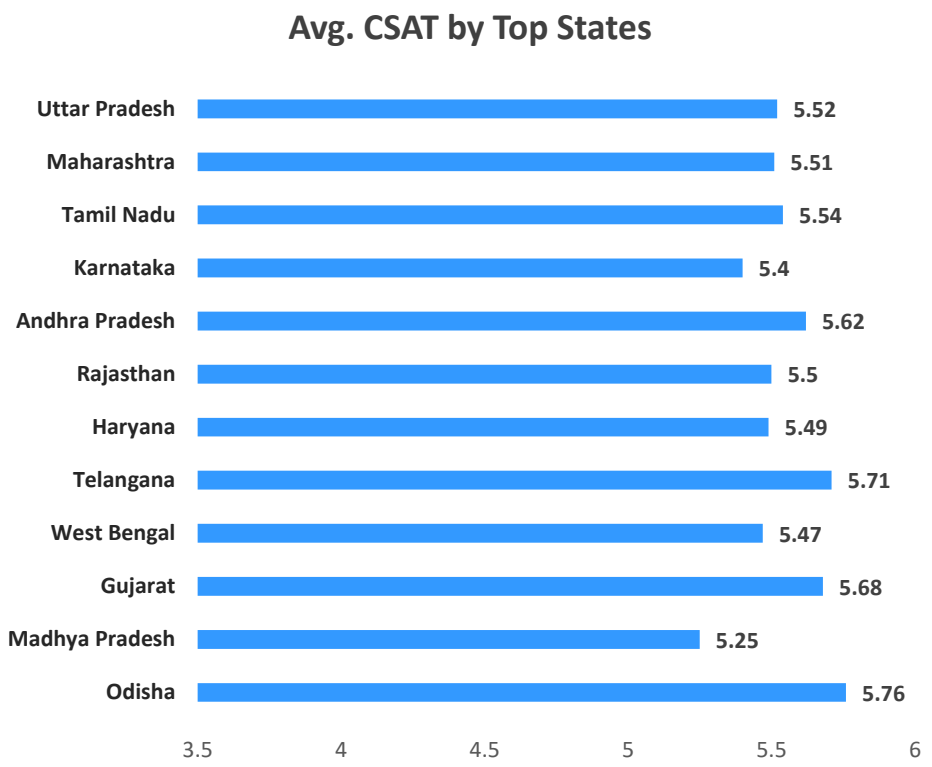
Insight:
Within each issue type, CSAT shows only minor variation across call centers with no consistent underperformance



Hypothesis:
Customer satisfaction (CSAT) varies significantly across top states.

Methodology:
- Segmented cases by customer state
- Calculated average CSAT score for each state
- Compared CSAT performance across top traffic states
- Visualized the comparison using a bar chart

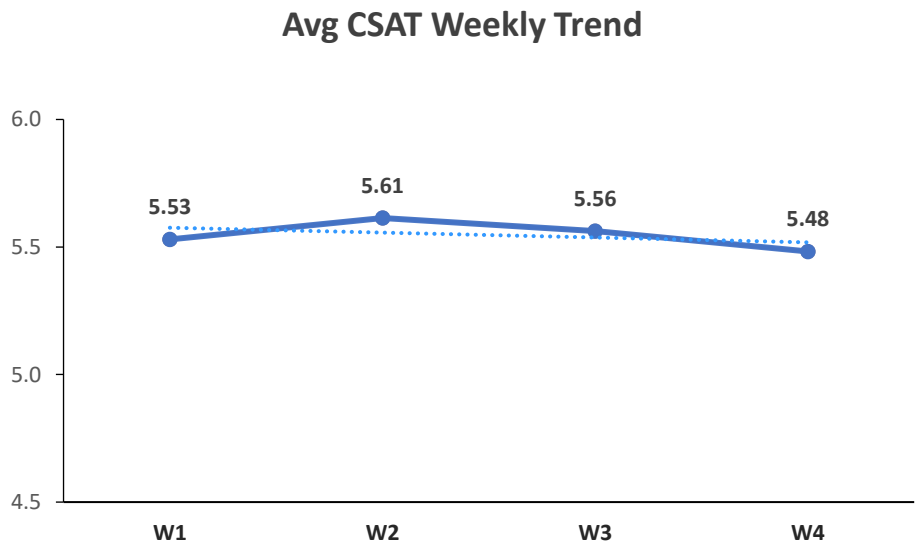
Insight:
While CSAT varies across states (5.25–5.76), the differences are relatively small, indicating no strong or consistent regional impact on customer satisfaction



Hypothesis:
CSAT scores decreased significantly throughout the month

Methodology:
- Segmented data by week (W1–W4)
- Calculated average CSAT score for each week
- Analyzed the trend of CSAT over time
- Visualized the pattern using a line chart

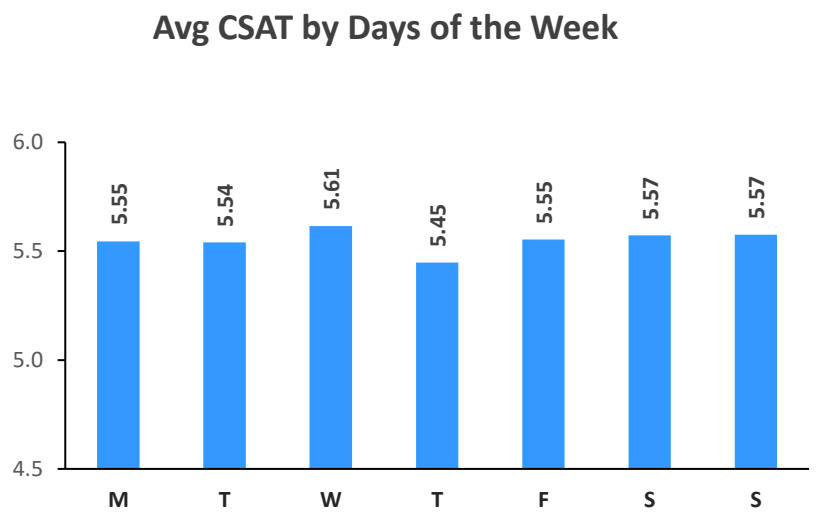
Insight:
CSAT scores show only minor fluctuations across weeks (5.48–5.61) with no significant downward trend, indicating that customer satisfaction remained relatively stable



Hypothesis:
Certain days of the week have lower CSAT scores.

Methodology:
- Segmented cases by day of the week (Mon–Sun)
- Calculated average CSAT score for each day
- Compared CSAT performance across days
- Visualized the comparison using a bar chart

Insight:
CSAT scores show minimal variation across days (5.45–5.61), with no specific day consistently underperforming

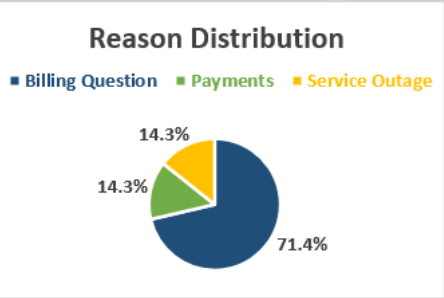


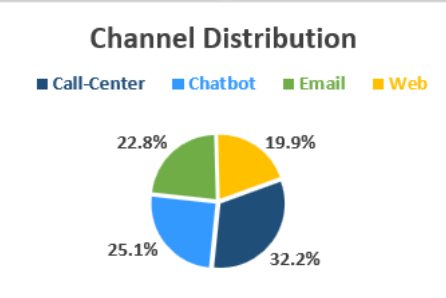
Flipkart Customer Support Analysis Dashboard | *Oct 2020*

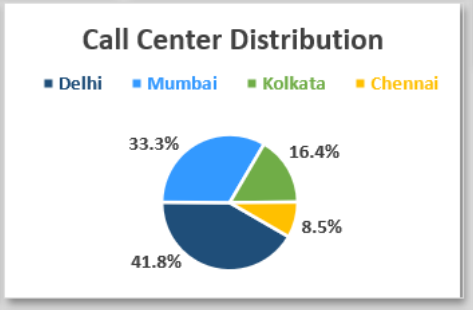
Reason
- Billing Question
- Payments
- Service Outage

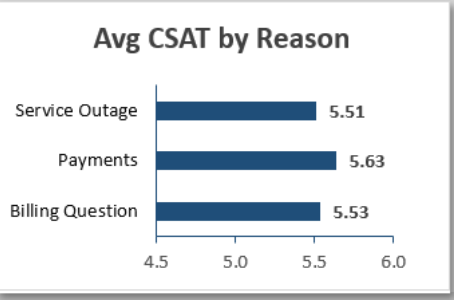
State
- Madhya Pr...
- Maharashtra
- Manipur
- Meghalaya
- Mizoram
- Nagaland
- Odisha
- Puducherry
- Punjab
- Rajasthan
- Sikkim

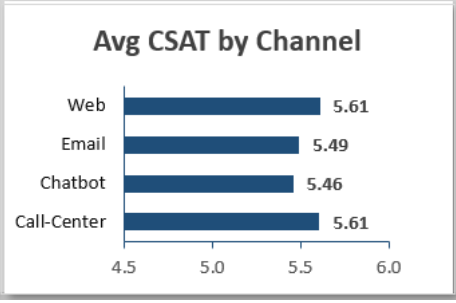
Total Interactions	Avg CSAT Score	SLA Breach Rate	Avg Call Duration
30K	5.54	12.7%	25 min

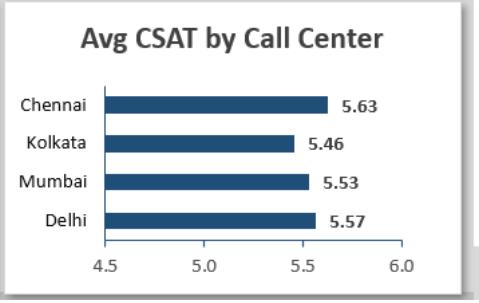


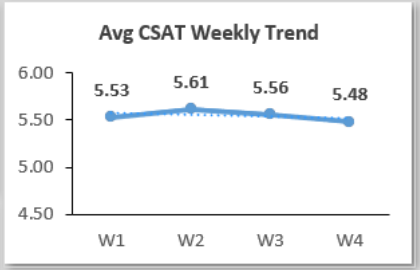


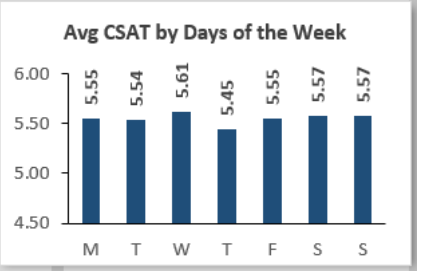












Channel Weekly CSAT Heatmap

	Week 1	Week 2	Week 3	Week 4
Call-Center	5.72	5.66	5.58	5.52
Chatbot	5.27	5.54	5.53	5.45
Email	5.42	5.59	5.58	5.34
Web	5.68	5.66	5.55	5.64

Call Center Weekly CSAT Heatmap

	Week 1	Week 2	Week 3	Week 4
Chennai	5.73	5.78	5.54	5.44
Delhi	5.52	5.65	5.59	5.54
Kolkata	5.32	5.39	5.68	5.40
Mumbai	5.60	5.63	5.48	5.47

Avg CSAT by Top States

Rank	State	Dist	CSAT
1	Uttar Pradesh	12.6%	5.52
2	Maharashtra	12.3%	5.51
3	Tamil Nadu	8.0%	5.54
4	Karnataka	7.4%	5.40
5	Andhra Pradesh	6.8%	5.62
6	Rajasthan	5.5%	5.50
7	Haryana	5.2%	5.49
8	Telangana	4.3%	5.71
9	West Bengal	4.2%	5.47
10	Gujarat	3.8%	5.68
11	Madhya Pradesh	3.3%	5.25
12	Odisha	3.2%	5.76

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Prioritize billing issues (~70% of interactions) by enhancing resolution quality and agent training in high-volume centers.

Customers prefer to reach out to call centers directly for payment queries; focus on faster diagnosis, clear transaction communication, and specialized support teams.

Service outage queries are mostly initiated via chatbot (~50%); enhance it with real-time updates, clear timelines, and seamless escalation to call centers when needed.

Delhi and Mumbai handle ~75% of interactions; focus on quality monitoring, standardization, and performance benchmarking to improve CSAT.

Since SLA and call duration have minimal impact, focus on interaction quality and resolution effectiveness, supported by sentiment tracking and quality audits.

Sentiment is the strongest driver of CSAT; use it to monitor performance, detect issues early, and enable proactive improvements through real-time tracking.

THANK YOU!

Flipkart Wholesale

Business Banaye Aasan

Shashwat Singh